

LISTA 6

1-a) $\|\vec{u}\| = \sqrt{1^2 + j^2 + k^2} = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$

b) $\|\vec{u}\| = \sqrt{3^2 + 0^2 + 9^2} = \sqrt{9 + 81} = \sqrt{90} = 3\sqrt{10}$

c) $\|\vec{u}\| = \sqrt{(-1)^2 + 1^2 + 0^2} = \sqrt{2}$

d) $\|\vec{u}\| = \sqrt{4^2 + 3^2 + (-1)^2} = \sqrt{16 + 9 + 1} = \sqrt{26}$

2-a) Supondo que $D = (0, 0, 0)$, então:

$H = (1, 0, 0) (\vec{DH})$, $C = (0, 1, 0) (\vec{DC})$, $A = (0, 0, 1) (\vec{DA})$

então:

$e_1 = \vec{DH} = H - D = (1 - 0, 0 - 0, 0 - 0) = (1, 0, 0)$

$e_2 = \vec{DC} = C - D = (0 - 0, 1 - 0, 0 - 0) = (0, 1, 0)$

$e_3 = \vec{DA} = A - D = (0 - 0, 0 - 0, 1 - 0) = (0, 0, 1)$

$B = A + C = (0, 1, 1)$ $F = C + H = (1, 1, 0)$

$E = A + H = (1, 0, 1)$ $G = A + H + C = (1, 1, 1)$

→ Para vetores ortogonais:

$$\left. \begin{aligned} \vec{e}_1 \cdot \vec{e}_2 &= (1, 0, 0) \cdot (0, 1, 0) = 0 \\ \vec{e}_1 \cdot \vec{e}_3 &= (1, 0, 0) \cdot (0, 0, 1) = 0 \\ \vec{e}_2 \cdot \vec{e}_3 &= (0, 1, 0) \cdot (0, 0, 1) = 0 \end{aligned} \right\} \begin{array}{l} \text{Toda as combinações} \\ \text{é igual a zero} \end{array}$$

• Toda os produtos entre os pares distintos são igual zero, dessa forma os vetores são ortogonais

Norma:

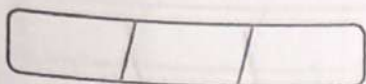
$\|\vec{e}_1\| = 1$

$\|\vec{e}_2\| = 1$

$\|\vec{e}_3\| = 1$

• O conjunto $E = (e_1, e_2, e_3)$ é uma base ortogonal, pois é LI e os produtos são zero, logo, ortogonal





b) $\vec{u}, \vec{v}, \vec{w}$ na base E

$$\vec{u} = c\vec{D} + c\vec{B} = 1$$

$$\vec{u} = (0 - 1) + (0 - 1)$$

$$\vec{u} = (0, -1, 0) + (0, 0, 1)$$

$$\vec{u} = (0, -1, 1)$$

$$\vec{v} = 5\vec{C} + c\vec{B}$$

$$\vec{v} = (0 - 1) + (0 - 1)$$

$$\vec{v} = (0, 1, 0) + (0, 0, 1)$$

$$\vec{v} = (0, 1, 1)$$

$$\vec{w} = 6\vec{C} = c - 6 = (-1, 0, 0)$$

c) $F = (\vec{f}_1, \vec{f}_2, \vec{f}_3)$ $\vec{f}_1 = \frac{\vec{u}}{\|\vec{u}\|}$, $\vec{f}_2 = \frac{\vec{v}}{\|\vec{v}\|}$, $\vec{f}_3 = \vec{w}$

$$\vec{u} = (0, -1, 1) \quad \vec{v} = (0, 1, 1) \quad \vec{w} = (-1, 0, 0)$$

$$\|\vec{u}\| = \sqrt{1^2 + (-1)^2 + 0^2} = \sqrt{2}$$

$$\|\vec{v}\| = \sqrt{1^2 + 0^2 + 1^2} = \sqrt{2} \quad \|\vec{w}\| = 1$$

Assim:

$$\vec{f}_1 = \frac{(0, -1, 1)}{\sqrt{2}} = \left(0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

$$\vec{f}_2 = \frac{(0, 1, 1)}{\sqrt{2}} = \left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

$$\vec{f}_3 = (-1, 0, 0)$$

ortogonais:

$$\vec{f}_1 \cdot \vec{f}_2 = \left(0 \cdot \frac{1}{\sqrt{2}} + \left(-\frac{1}{\sqrt{2}}\right) \cdot \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}\right) = 0$$

$$\vec{f}_1 \cdot \vec{f}_3 = \left(0 \cdot (-1) + \left(-\frac{1}{\sqrt{2}}\right) \cdot 0 + \frac{1}{\sqrt{2}} \cdot 0\right) = 0$$

$$\vec{f}_2 \cdot \vec{f}_3 = \left(0 \cdot (-1) + \frac{1}{\sqrt{2}} \cdot 0 + \frac{1}{\sqrt{2}} \cdot 0\right) = 0$$

ortogonais

$\therefore F(\vec{f}_1, \vec{f}_2, \vec{f}_3)$ é uma base ortogonal

d) $(ME \rightarrow F)$

$$\vec{f}_1 = (0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}) \in$$

$$\vec{f}_2 = (0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}) \in$$

$$\vec{f}_3 = (-1, 0, 0)$$

$$M = \begin{pmatrix} 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -1 & 0 & 0 \end{pmatrix}$$



e) $\vec{HB} = B(0, 1, 1) - H(1, 0, 0)$

$$B-H = (0, 1, 1) - (1, 0, 0) = (0-1, 1-0, 1-0) = (-1, 1, 1)$$

$$MT \rightarrow C = \begin{pmatrix} 0 & -1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} = (0, \sqrt{2}, 1)$$

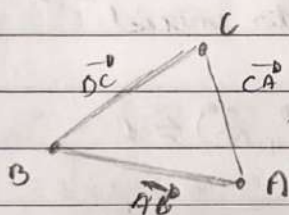
3a) $A = (2, 4, 3), B = (5, 1, -3), C = (0, -3, 1)$

Vetores:

$$\vec{AB} = B - A = (5-2, 1-4, -3-3) = (3, -3, -6)$$

$$\vec{BC} = C - B = (0-5, -3-1, 1-(-3)) = (-5, -4, 4)$$

$$\vec{CA} = A - C = (2-0, 4-(-3), 3-1) = (2, 7, 2)$$



3)

$$\|\vec{AB}\| = \sqrt{3^2 + (-3)^2 + (-6)^2} = \sqrt{54} = 3\sqrt{6}$$

$$\|\vec{BC}\| = \sqrt{(-5)^2 + (-4)^2 + 4^2} = \sqrt{57}$$

$$\|\vec{CA}\| = \sqrt{2^2 + 7^2 + 2^2} = \sqrt{57}$$

c) $M_{AB} = \left(\frac{5+2}{2}, \frac{1+4}{2}, \frac{-3+3}{2} \right) = \left(\frac{7}{2}, \frac{5}{2}, 0 \right)$

$$M_{BC} = \left(\frac{0+5}{2}, \frac{-3+1}{2}, \frac{1-3}{2} \right) = \left(\frac{5}{2}, -1, -1 \right)$$

$$M_{CA} = \left(\frac{2+0}{2}, \frac{4-3}{2}, \frac{3+1}{2} \right) = \left(1, \frac{1}{2}, 2 \right)$$

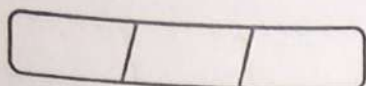
Mediana relativa AB:

$$\vec{CM_{AB}} = M_{AB} - C = \left(\frac{7}{2} - 0, \frac{5}{2} - (-3), 0 - 1 \right) = \left(\frac{7}{2}, \frac{11}{2}, -1 \right)$$

Mediatriz: $\vec{AB} \cdot \vec{CM_{AB}} = 3 \cdot \frac{7}{2} + (-3) \cdot \frac{11}{2} + (-6) \cdot (-1) = \frac{21}{2} - \frac{33}{2} + 6 = 0$

• produto geo) AB coincide com a mediatriz





d) $\hat{\text{ângulo}} \ B \hat{C} \ A$

$$\vec{CB} \cdot \vec{CA}$$

$$(5, 4, -2) \cdot (2, 7, 2)$$

$$(5 \cdot 2 + 4 \cdot 7 + (-2) \cdot 2)$$

$$10 + 28 - 4$$

$$30$$

$$\vec{CB} = B - C = (5-0, 1+5, -3+1) = (5, 6, -2)$$

$$\|\vec{CB}\| = \sqrt{5^2 + 6^2 + (-2)^2} = \sqrt{57}$$

$$\cos \theta = \frac{\vec{CB} \cdot \vec{CA}}{\|\vec{CB}\| \cdot \|\vec{CA}\|} = \frac{30}{\sqrt{57} \cdot \sqrt{57}} = \frac{30}{57} = 0,526 \approx 58,2^\circ$$

e) $\vec{AB} + \vec{BC} + \vec{CA}$

$$B - A + C - B + A - C = (0, 0, 0)$$

A soma é zero porque os vetores formam um ciclo fechado (triângulo) volta para o ponto inicial

4-a) $|\vec{u} \cdot \vec{v}| \leq \|\vec{u}\| \cdot \|\vec{v}\|$ $\cos(\vec{u}, \vec{v}) \leq 1$

$$\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cdot \cos(\vec{u}, \vec{v})$$

$$|\vec{u} \cdot \vec{v}| \leq \|\vec{u}\| \|\vec{v}\|$$

3) $\|\vec{u} + \vec{v}\| \leq \|\vec{u}\| + \|\vec{v}\|$

$$\|\vec{u} + \vec{v}\|^2 = \|\vec{u}\|^2 + 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2$$

$$\|\vec{u} + \vec{v}\|^2 \leq \|\vec{u}\|^2 + 2\|\vec{u}\|\|\vec{v}\| + \|\vec{v}\|^2$$

$$\sqrt{\|\vec{u} + \vec{v}\|^2} \leq \sqrt{\|\vec{u}\|^2 + 2\|\vec{u}\|\|\vec{v}\| + \|\vec{v}\|^2}$$

$$\|\vec{u} + \vec{v}\| \leq \|\vec{u}\| + \|\vec{v}\|$$

c) $\|\vec{u} + \vec{v}\|^2 = \|\vec{u} - \vec{v}\|^2$

$$\|\vec{u}\|^2 + 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2 = (\|\vec{u}\|^2 - 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2) \Rightarrow 4\vec{u} \cdot \vec{v}$$

$$\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2 = 4\vec{u} \cdot \vec{v}$$

5 a) $\vec{u} = (1, 0, 1)$, $\vec{v} = (-2, 10, 2)$

$$\vec{u} \cdot \vec{v} = 1 \cdot (-2) + 0 \cdot 10 + 1 \cdot 2 = 0$$

$$\|\vec{u}\| = \sqrt{1^2 + 0^2 + 1^2} = \sqrt{2}$$

$$\|\vec{v}\| = \sqrt{2^2 + 10^2 + 2^2} = \sqrt{108} = 6\sqrt{3}$$

$$\cos \theta = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \|\vec{v}\|}$$

$$\cos \theta = \frac{0}{\sqrt{2} \cdot 6\sqrt{3}}$$

$$\cos \theta = 0 \Rightarrow \theta = \frac{\pi}{2} \text{ radianos}$$

$$90^\circ$$

$$b) \vec{u}^0 = (-1, 1, 1), \vec{v}^0 = (1, 1, 1)$$

$$\vec{u}^0 \cdot \vec{v}^0 = (-1 \cdot 1 + 1 \cdot 1 + 1 \cdot 1) = 1$$

$$\cos \theta = \frac{1}{\sqrt{3} \cdot \sqrt{3}} = \frac{1}{3} \approx 1,23 \text{ rad}$$

$$\|\vec{u}^0\| = \sqrt{(-1)^2 + 1^2 + 1^2} = \sqrt{3}$$

$$\|\vec{v}^0\| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

$$c) \vec{u}^0 = (3, 3, 0), \vec{v}^0 = (2, 1, -2)$$

$$\vec{u}^0 \cdot \vec{v}^0 = (3 \cdot 2 + 3 \cdot 1 + 0 \cdot (-2)) = 9$$

$$\cos \theta = \frac{9}{3\sqrt{3} \cdot 3} = \frac{1}{\sqrt{3}} = 60^\circ = \frac{\pi}{3} \text{ rad}$$

$$\|\vec{u}^0\| = \sqrt{3^2 + 3^2 + 0} = \sqrt{18} = 3\sqrt{2}$$

$$\|\vec{v}^0\| = \sqrt{2^2 + 1^2 + (-2)^2} = \sqrt{9} = 3$$

$$d) \vec{u}^0 = (\sqrt{3}, 1, 0), \vec{v}^0 = (\sqrt{3}, 1, 2\sqrt{3})$$

$$\vec{u}^0 \cdot \vec{v}^0 = (\sqrt{3} \cdot \sqrt{3} + 1 \cdot 1 + 0 \cdot 2\sqrt{3}) = 4$$

$$\cos \theta = \frac{4}{2 \cdot 4} = \frac{1}{2} = 60^\circ = \frac{\pi}{3} \text{ rad}$$

$$\|\vec{u}^0\| = \sqrt{(\sqrt{3})^2 + 1^2 + 0} = \sqrt{4} = 2$$

$$\|\vec{v}^0\| = \sqrt{(\sqrt{3})^2 + 1^2 + (2\sqrt{3})^2} = \sqrt{16} = 4$$

$$6-a) \vec{u}^0 = (x+1, 1, 2), \vec{v}^0 = (x-1, -1, -2)$$

$$\vec{u}^0 \cdot \vec{v}^0 = (x+1) \cdot (x-1) + 1 \cdot (-1) + 2 \cdot (-2) = 0$$

$$x^2 - 1 - 1 - 4 = 0$$

$$x^2 - 6 = 0$$

$$x^2 = 6 \quad \begin{cases} x_1 = +\sqrt{6} \\ x_2 = -\sqrt{6} \end{cases}$$

$$b) \vec{u}^0 = (x, x, 4), \vec{v}^0 = (4, x, 1)$$

$$\vec{u}^0 \cdot \vec{v}^0 = (x \cdot 4 + x \cdot x + 4 \cdot 1) = 0$$

$$x^2 + 4x + 4 = 0$$

$$\Delta = 4^2 - 4 \cdot 1 \cdot 4 \quad x = \frac{-4}{2} = -2$$

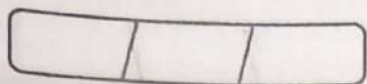
$$\Delta = 0$$

7a)

x	y	z
4	-1	5
1	-2	3

$$= -3x - 8z + 5y + z + 10x - 12y$$

$$-7x - 7z - 7y \Rightarrow (7, -7, -7)$$



$$(1, -1, -1) \cdot (1, 1, 1) = 1 - 1 - 1 = -1$$

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$$(7, -7, -7)$$

$$b) \begin{pmatrix} x & y & z \\ 2 & 3 & -1 \\ 2 & -4 & 6 \end{pmatrix} = \begin{pmatrix} 18x - 8z - 2y - 6y - 4x - 12y \\ 14x - 14y - 14z \end{pmatrix}$$

$$(14, -14, -14)$$

$\hookrightarrow$ paralelo a $(1, -1, -1)$

$$\text{Norma: } \vec{u} = \frac{1}{\sqrt{3}} \cdot (1, -1, -1) = (3, -3, -3)$$

$$\vec{u} \cdot \vec{j} = 3 > 0 \text{ (ângulo agudo)}$$

$$\text{oposto } \vec{u} \cdot \vec{i} = -3 < 0 \text{ (ângulo obtuso)}$$

$$c) \vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos\left(\frac{\pi}{4}\right) = 5 \cdot 1 \cdot \frac{\sqrt{2}}{2} = \frac{5\sqrt{2}}{2}$$

$$\vec{u} + \vec{v} \Rightarrow \|\vec{u} + \vec{v}\|^2 = \|\vec{u}\|^2 + 2 \cdot \vec{u} \cdot \vec{v} + \|\vec{v}\|^2 = 26 + 5\sqrt{2} = 24$$

$$\vec{u} - \vec{v} \Rightarrow \|\vec{u} - \vec{v}\|^2 = \|\vec{u}\|^2 - 2 \cdot \vec{u} \cdot \vec{v} + \|\vec{v}\|^2 = 26 - 5\sqrt{2}$$

$$(\vec{u} + \vec{v}) \cdot (\vec{u} - \vec{v})$$

$$\|\vec{u}\|^2 - \|\vec{v}\|^2 = 25 - 1 = 24$$

$$\theta = \cos \theta = \frac{24}{\sqrt{26+5\sqrt{2}} \cdot \sqrt{26-5\sqrt{2}}} = \frac{24}{\sqrt{26^2 - (5\sqrt{2})^2}} = \frac{24}{\sqrt{626}} \Rightarrow \theta_{\text{arccos}} = 24$$

$$8 - \vec{v} = (1, -1, 2) \quad \vec{u} = (3, -1, 1)$$

$$\text{Proj} \left(\frac{\vec{u} \cdot \vec{v}}{\vec{u} \cdot \vec{u}} \right) \vec{u}$$

a)

$$\vec{u} \cdot \vec{v} = (3 \cdot 1 + (-1) \cdot (-1) + 1 \cdot 2) = 6$$

$$\vec{u} \cdot \vec{u} = (3 \cdot 3 + (-1) \cdot (-1) + 1 \cdot 1) = 11$$

$$\text{Proj} = \left(\frac{6}{11} \right) \cdot (3, -1, 1) = \left(\frac{18}{11}, -\frac{6}{11}, \frac{6}{11} \right)$$



$$b) \vec{v} = (1, 3, 5), \vec{u} = (-3, 1, 0)$$

$$\vec{u} \cdot \vec{v} = (-3) \cdot 1 + 1 \cdot 3 + 0 \cdot 5 = 0$$

$$\text{projecção} = (0, 0, 0)$$

$$c) \vec{v} = (-1, 1, 1), \vec{u} = (-2, 1, 2)$$

$$\vec{u} \cdot \vec{v} = (-2) \cdot (-1) + 1 \cdot 1 + 1 \cdot 1 = 5$$

$$|\vec{u}| = \sqrt{(-2)^2 + 1^2 + 2^2} = 3$$

$$\text{projecção} = \frac{5}{9} \cdot (-2, 1, 2) = \left(-\frac{10}{9}, \frac{5}{9}, \frac{10}{9}\right)$$

$$d) \vec{v} = (1, 2, 4), \vec{u} = (-2, -4, -8)$$

$$\vec{u} \cdot \vec{v} = (-2) \cdot 1 + (-4) \cdot 2 + (-8) \cdot 4 = -42$$

$$|\vec{u}| = \sqrt{(-2)^2 + (-4)^2 + (-8)^2} = 8.94$$

$$\text{projecção} = \frac{-42}{84} \cdot (-2, -4, -8) = \frac{1}{2} \cdot (-2, -4, -8) = (-1, -2, -4)$$

$$9-a) \vec{u} = (2, -2, 1), \vec{v} = (3, -6, 0)$$

$$\vec{u} \cdot \vec{v} = (2 \cdot 3 + (-2) \cdot (-6) + 1 \cdot 0) = 18$$

$$|\vec{u}| = \sqrt{2^2 + (-2)^2 + 1^2} = 3$$

$$\text{proj} = \frac{18}{9} \cdot (2, -2, 1) = 2 \cdot (2, -2, 1) = (4, -4, 2)$$

$$\text{projecção} = \frac{18}{45} \cdot (3, -6, 0) = \left(\frac{6}{5}, -\frac{12}{5}, 0\right)$$

$$|\vec{v}| = \sqrt{3^2 + (-6)^2 + 0^2} = 6.71$$

$$b) \vec{N} = \vec{P} + \vec{Q}$$

$\vec{P}$ é paralelo a $\vec{u}$: $\vec{P} = \rho \vec{u}$ $\vec{N} = (4, -4, 2)$

$\vec{Q}$ é ortogonal a $\vec{u}$: $\vec{Q} = \vec{N} - \vec{P}$

$$\vec{Q} = (3, -6, 0) - (4, -4, 2) = (-1, -2, -2)$$

$$\text{ortogonal: } \vec{u} \cdot \vec{Q} = 2(-1) + (-2)(-2) + 1(-2) = -2 + 4 - 2 = 0$$

$$\vec{P} = (4, -4, 2) \quad \vec{Q} = (-1, -2, -2)$$

$$c) \begin{array}{|c|c|c|} \hline \vec{u} \times \vec{v} = & x & y & z \\ \hline & 2 & -2 & 1 \\ \hline & 3 & -6 & 0 \\ \hline \end{array} = 6x + 3y - 6z$$

$$(6, 3, -6)$$

area:

$$\|\vec{u} \times \vec{v}\| = \sqrt{6^2 + 3^2 + (-6)^2} = \sqrt{81} = 9$$

$$10 - a) \vec{u} = 3\vec{i} + 3\vec{j} \quad \vec{v} = 5\vec{i} + 4\vec{j}$$

$$\vec{u} = (3, 3, 0) \quad \vec{v} = (5, 4, 0)$$

$$\vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 3 & 0 \\ 5 & 4 & 0 \end{vmatrix} = 0\vec{i} + 0\vec{j} - 3\vec{k}$$

$$(0, 0, -3)$$

$$\|\vec{u} \times \vec{v}\| = \sqrt{0^2 + 0^2 + (-3)^2} = \sqrt{9} = 3$$

$$b) \vec{u} = (7, 0, -5) \quad \vec{v} = (1, 2, -1)$$

$$\vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 7 & 0 & -5 \\ 1 & 2 & -1 \end{vmatrix} = 10\vec{i} + 2\vec{j} + 14\vec{k}$$

$$(10, 2, 14)$$

$$\|\vec{u} \times \vec{v}\| = \sqrt{10^2 + 2^2 + 14^2} = \sqrt{300} = 10\sqrt{3}$$

$$c) \begin{array}{|c|c|c|} \hline \vec{r} \times \vec{s} & \vec{i} & \vec{j} & \vec{k} \\ \hline & 1 & -3 & 1 \\ \hline & 1 & 1 & 4 \\ \hline \end{array} = -13\vec{i} - 3\vec{j} + 4\vec{k}$$

$$(-13, -3, 4)$$

$$\|\vec{r} \times \vec{s}\| = \sqrt{(-13)^2 + (-3)^2 + 4^2} = \sqrt{194}$$

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$$d) \vec{u} = (2, 1, 2), \vec{v} = (1, 2, 1)$$

$$\vec{u} \times \vec{v} = \begin{vmatrix} i & j & k \\ 2 & 1 & 2 \\ 1 & 2 & 1 \end{vmatrix} = 2 \times 1 - 2 \times 2 = 0 \Rightarrow (0, 0, 0)$$

11-a) produto vetorial: $\|\vec{u} \times \vec{v}\| = \|\vec{u}\| \|\vec{v}\| \sin \theta$

produto escalar: $\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos \theta$

$$\begin{aligned} \|\vec{u}\|^2 \|\vec{v}\|^2 - (\vec{u} \cdot \vec{v})^2 &= \|\vec{u}\|^2 \|\vec{v}\|^2 - \|\vec{u}\|^4 \|\vec{v}\|^2 \cos^2 \theta \\ &= \|\vec{u}\|^2 \|\vec{v}\|^2 (1 - \cos^2 \theta) \\ &= \|\vec{u}\|^2 \|\vec{v}\|^2 \sin^2 \theta \end{aligned}$$

$$\|\vec{u} \times \vec{v}\|^2 = (\|\vec{u}\| \|\vec{v}\| \sin \theta)^2 = \|\vec{u}\|^2 \|\vec{v}\|^2 \sin^2 \theta$$

$$\|\vec{u} \times \vec{v}\|^2 = \|\vec{u}\|^4 \|\vec{v}\|^2 - (\vec{u} \cdot \vec{v})^2$$

b) $\|\vec{u} \times \vec{v}\| \quad \vec{u} \cdot \vec{v} = 3 \quad \|\vec{u}\| = 1 \quad \|\vec{v}\| = 5$

$$\|\vec{u} \times \vec{v}\|^2 = (1)^2 (5)^2 - (3)^2 = 25 - 9 = 16$$

$$\|\vec{u} \times \vec{v}\| = \sqrt{16} = 4$$

c) $\|\vec{AB} \times \vec{AC}\| \quad \vec{AB} \cdot \vec{AC} = \theta = 60^\circ$

$$\|\vec{AB} \times \vec{AC}\| = \|\vec{AB}\| \|\vec{AC}\| \sin \theta = l \cdot l \cdot \sin 60^\circ = l^2 \cdot \frac{\sqrt{3}}{2}$$

$$\text{Área do triângulo} = \frac{1}{2} \|\vec{AB} \times \vec{AC}\| = \frac{l^2 \sqrt{3}}{4}$$

$$\|\vec{AB} \times \vec{AC}\| = \frac{l^2 \sqrt{3}}{2}$$

$$12-a) \begin{cases} \vec{x} \cdot (2\vec{i} + 3\vec{j} + 4\vec{k}) = 9 \\ \vec{x} \cdot (-\vec{i} + \vec{j} - \vec{k}) = -2\vec{i} + 2\vec{k} \end{cases}$$

$$I \quad \vec{x} = a\vec{i} + b\vec{j} + c\vec{k} \Rightarrow \vec{x} \cdot (2\vec{i} + 3\vec{j} + 4\vec{k}) = 2a + 3b + 4c = 9$$

$$II \quad \vec{x} \cdot (-\vec{i} + \vec{j} - \vec{k}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a & b & c \\ -1 & 1 & -1 \end{vmatrix} = \vec{i}(b \cdot (-1) - c \cdot 1) - \vec{j}(a \cdot (-1) - c \cdot (-1)) + \vec{k}(a \cdot 1 - b \cdot (-1))$$

$$= \vec{i}(-b-c) - \vec{j}(-a+c) + \vec{k}(a+b)$$

$$= (-b-c)\vec{i} + (a-c)\vec{j} + (a+b)\vec{k}$$

$$(-b-c)\vec{i} + (a-c)\vec{j} + (a+b)\vec{k} = -2\vec{i} + 0\vec{j} + 2\vec{k}$$

$$\begin{cases} -b-c = -2 & I & 2a+3b+4c=9 & c=a=1 \\ a-c=0 & II & 2a+3(2-a)+4a=9 & b=2-a \\ a+b=2 & III & 3a+6=9 \Rightarrow a=1 & b=1 \end{cases}$$

$$S: \vec{x} = \vec{i} + \vec{j} + \vec{k}$$

$$b) \begin{cases} \vec{x} \cdot (1, 0, 1) = 2(1, 1, -1) \\ \|\vec{x}\| = 6 \end{cases}$$

$$I \quad \vec{x} = (a, b, c) \quad \vec{x} \cdot (1, 0, 1) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a & b & c \\ 1 & 0 & 1 \end{vmatrix} = b\vec{i} - (a-c)\vec{j} - b\vec{k}$$

$$b\vec{i} - (a-c)\vec{j} - b\vec{k} = 2\vec{i} + 2\vec{j} - 2\vec{k}$$

$$b=2$$

$$a-c=-2$$

$$\|\vec{x}\| = \sqrt{a^2 + b^2 + c^2} = 6 \Rightarrow a^2 + b^2 + c^2 = 36$$

$$a^2 + 4 + c^2 = 36$$

$$a^2 + c^2 = 32$$

$$\begin{cases} a^2 + c^2 = 32 \Rightarrow a = -1 \pm \sqrt{15} \\ a-c = -2 \end{cases}$$

$$c = 1 \pm \sqrt{15}$$

$$\text{FORONI } S: \vec{x} = (-1 + \sqrt{15}, 2, 1 + \sqrt{15}) \text{ ou } \vec{x} = (-1 - \sqrt{15}, 2, 1 - \sqrt{15})$$



$$c) \|\vec{x}\| = 3$$

$\vec{x}$ é ortogonal a $\vec{u} = (-3, 0, 3)$ e $\vec{v} = (2, -2, 0)$

$\vec{x}$ forma ângulo obtuso com $\vec{j}$

Ortogonalidade: $\vec{x} \cdot \vec{u} = -3a + 0b + 3c = 0 \Rightarrow -3a + 3c = 0 \quad a = c$

$\vec{x} \cdot \vec{v} = 2a - 2b + 0c = 0 \Rightarrow 2a - 2b = 0 \quad a = b$

assim: $a = b = c$

$$\|\vec{x}\| = \sqrt{a^2 + b^2 + c^2} = 3 \Rightarrow \sqrt{3a^2} = 3 \Rightarrow a\sqrt{3} = 3$$

$$a = \pm\sqrt{3}$$

assim: $\vec{x} = (\sqrt{3}, \sqrt{3}, \sqrt{3})$ ou $\vec{x} = (-\sqrt{3}, -\sqrt{3}, -\sqrt{3})$

ângulo θ entre $\vec{x}$ e $\vec{j}$ satisfaz $\cos \theta = \frac{\vec{x} \cdot \vec{j}}{\|\vec{x}\| \|\vec{j}\|} = \frac{b}{3}$

para ser obtuso, $\cos \theta < 0, b < 0$

o que forma: $\vec{x} = (-\sqrt{3}, -\sqrt{3}, -\sqrt{3})$

13-a) Área: ABCD, $\vec{AB} = (1, 1, -1)$, $A = (3, 2, -1)$ e $D = (5, 3, 3)$

$$\vec{AD} = D - A = (5 - 3, 3 - 2, 3 + 1) = (2, 1, 4)$$

$$\vec{AB} \times \vec{AD} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & -1 \\ 2 & 1 & 4 \end{vmatrix} = (4+1)\vec{i} - (4+2)\vec{j} + \vec{k}(1-2) = 5\vec{i} - 6\vec{j} - \vec{k}$$

$$\|\vec{AB} \times \vec{AD}\| = \sqrt{5^2 + (-6)^2 + (-1)^2} = \sqrt{25 + 36 + 1} = \sqrt{62}$$

ABCD é $\sqrt{62}$

b) Vetor $\vec{AB} = (-1, 1, 0)$

Vetor $\vec{AC} = (0, 1, 3)$

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -1 & 1 & 0 \\ 0 & 1 & 3 \end{vmatrix} = 3\vec{i} + 3\vec{j} - \vec{k}$$

$$\|\vec{AB} \times \vec{AC}\| = \sqrt{3^2 + 3^2 + (-1)^2} = \sqrt{9 + 9 + 1} = \sqrt{19}$$

Área = $\frac{1}{2} \|\vec{AB} \times \vec{AC}\| = \frac{\sqrt{19}}{2}$

$$h_A = \frac{2 \times \text{Area}}{\|\vec{BC}\|}$$

$$\vec{BC} = \vec{AC} - \vec{AB} = (0+1, 1-1, 3-0)$$

$$\vec{BC} = (1, 0, 3)$$

$$h_A = \frac{2 \times \frac{\sqrt{19}}{2}}{\sqrt{10}} = \frac{\sqrt{19}}{\sqrt{10}}$$

$$\|\vec{BC}\| = \sqrt{1^2 + 0^2 + 3^2} = \sqrt{10}$$

$$\therefore \text{Altura relativa: } \sqrt{19/10}$$

$$14a) \begin{array}{l} \vec{u} = (x_1, y_1, z_1) \\ \vec{v} = (x_2, y_2, z_2) \\ \vec{w} = (x_3, y_3, z_3) \end{array} \quad \vec{v} \times \vec{w} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = \vec{i}(y_2 z_3 - z_2 y_3) - \vec{j}(x_2 z_3 - z_2 x_3) + \vec{k}(x_2 y_3 - y_2 x_3)$$

$$\vec{u} \cdot (\vec{v} \times \vec{w}) = x_1(y_2 z_3 - z_2 y_3) - y_1(x_2 z_3 - z_2 x_3) + z_1(x_2 y_3 - y_2 x_3)$$

$$\vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix} = \vec{i}(y_1 z_2 - z_1 y_2) - \vec{j}(x_1 z_2 - z_1 x_2) + \vec{k}(x_1 y_2 - y_1 x_2)$$

$$(\vec{u} \times \vec{v}) \cdot \vec{w} = x_3(y_1 z_2 - z_1 y_2) - y_3(x_1 z_2 - z_1 x_2) + z_3(x_1 y_2 - y_1 x_2)$$

$$\text{Dessa forma: } \vec{u} \cdot (\vec{v} \times \vec{w}) = (\vec{u} \times \vec{v}) \cdot \vec{w} = \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}$$

$$b) [\vec{u}, \vec{v}, \vec{w}] = \begin{vmatrix} 1 & 3 & 2 \\ 0 & 1 & -2 \\ -1 & 2 & 0 \end{vmatrix} = 4 + 6 + 2 = 12$$

$$[\vec{u}, \vec{v}, \vec{w}] = -[\vec{u}, \vec{v}, \vec{w}] = -12$$

$$[\vec{v}, 2\vec{w}, \vec{u}] \Rightarrow 2\vec{w} = (-2, 4, 0)$$

$$2 \cdot [\vec{v}, \vec{w}, \vec{u}] = 2(-[\vec{u}, \vec{v}, \vec{w}]) = 2 \cdot (-12) = -24$$

$$[\vec{u}, 3\vec{v} - 2\vec{w}, \vec{u} + 5\vec{w}] = 36$$

$$3\vec{v} - 2\vec{w} = (0, 3, -6) - (-2, 4, 0) = (-2, -3, -6)$$

$$\text{FORONI: } \vec{u} + 3\vec{w} = (-1, 2, 0) + (3, 9, 6) = (2, 11, 6)$$

$$\begin{vmatrix} 1 & 3 & 2 \\ -2 & -3 & -6 \\ 2 & 11 & 6 \end{vmatrix} = 36$$

15-c) $ABCD \rightarrow \vec{AB} \text{ e } \vec{AD}$

$$\vec{AF} = \vec{AB} + \vec{AD} + \vec{AE} \Rightarrow \vec{AD} = \vec{AF} - \vec{AB} - \vec{AE}$$

$$\vec{AD} = (3, 5, 6) - (1, 0, 1) - (3, 2, 3) = (-1, 3, 2)$$

$$\vec{AB} \times \vec{AD} = \begin{vmatrix} i & j & k \\ 1 & 0 & 1 \\ -1 & 3 & 2 \end{vmatrix} = -3i - 3j + 3k$$

$$(-3, -3, 3)$$

$$\text{Área} = \|\vec{AB} \times \vec{AD}\| = \sqrt{(-3)^2 + (-3)^2 + (3)^2} = \sqrt{27} = 3\sqrt{3}$$

5) Volume $[\vec{AB}, \vec{AD}, \vec{AE}]$

$$\begin{vmatrix} 1 & 0 & 1 \\ -1 & 3 & 2 \\ 3 & 2 & 3 \end{vmatrix} = 5 - 11 = -6 \quad \text{Volume} : |-6| = 6$$

c) Altura

$$h = \frac{\text{Volume}}{\text{área base}} = \frac{6}{3\sqrt{3}} = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$$

d) Volume da DD = $\frac{1}{6}$ total

$$\vec{EA} = -\vec{AE} = (-3, -2, -3)$$

$$\vec{EB} = \vec{EB} = \vec{EA} + \vec{AB} = (-3, -2, -3) + (1, 0, 1) = (-2, -2, -2)$$

$$\vec{ED} = \vec{EA} + \vec{AD} = (-3, -2, -3) + (-1, 3, 2) = (-4, 1, -1)$$

$$[\vec{EA}, \vec{EB}, \vec{ED}] = \begin{vmatrix} -3 & -2 & -3 \\ -2 & -2 & -2 \\ -4 & 1 & -1 \end{vmatrix} = -12 - 12 + 30 = 6$$

$$\text{Volume do tetraedro} = \frac{1}{6} \times 6 = 1$$

$$\|\vec{DE} \times \vec{DB}\| = \sqrt{(4^2 + 6^2 + 10^2)} = \sqrt{152}$$

e) Altura

$$\vec{DE} = -\vec{ED} = (4, -1, 1)$$

$$\vec{DB} = \vec{DE} + \vec{EB} = (4, -1, 1) + (-2, -2, -2) = (2, -3, -1)$$

$$\text{Área} = \frac{1}{2} \sqrt{152} = \frac{2\sqrt{38}}{2} = \sqrt{38} \quad \therefore h = \frac{3\sqrt{38}}{38}$$

$$\vec{DE} \times \vec{DB} = \begin{vmatrix} i & j & k \\ 4 & -1 & 1 \\ 2 & -3 & -1 \end{vmatrix} = 4i + 6j - 10k$$

$$= (4, 6, -10)$$

FORONI